

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWUID2026217449
Project Title	Rising Waters - Flood Prediction ML
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to build a machine learning-powered flood prediction system that analyzes weather indicators to forecast flood risk in advance, enabling timely evacuation and disaster response planning.
Scope	The project covers data collection, preprocessing, training and comparing multiple ML classification models, and deploying the best-performing model through a Flask web application for real-time flood risk prediction.
Problem Statement	
Description	Conventional flood forecasting methods often fail to provide timely and accurate warnings, leaving communities and authorities with insufficient time to respond, resulting in avoidable loss of life and property.
Impact	Solving this issue will result in earlier flood warnings, better-prepared disaster response teams, and reduced loss of life and property during flood events.
Proposed Solution	
Approach	Employing machine learning classification techniques (Decision Tree, Random Forest, KNN, XGBoost) trained on weather data including annual rainfall, cloud visibility, temperature, humidity, and seasonal rainfall to predict flood likelihood.
Key Features	- Multi-model comparison to select the most accurate classifier - Real-time prediction via a Flask web interface - Simple, accessible input form for weather indicators - Model and scaler persistence for fast repeated predictions

	- Real-time flood risk classification (High/Low) for quicker disaster response decisions - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (no GPU required)
Memory	RAM specifications	4 GB
Storage	Disk space for data, models, and logs	2 GB free space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, xgboost, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Synthetically generated weather dataset, 2000 records, CSV format